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Transformer-based deep learning model and video dataset for installation action recognition in offsite projects

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ABSTRACT

This paper developed and evaluated the Precast Concrete Installation Dataset (PCI-Dataset), a large-scale video dataset for automatically recognizing precast concrete (PC) installation activities. The dataset comprises 12,791 video clips (5 s each, 1080 × 1080 resolution, 30fps) from actual PC construction sites, including 12 balanced activity classes combining three component types and four work stages. Evaluation of six Transformer-based video classification models showed VideoMAE V2 achieved the highest overall accuracy of 98.10 %, followed by UniFormer V2, Video Swin, MVIT, ViViT, and TimeSformer. VideoMAE V2 achieved F1 scores above 80 % for most activities, with a peak of 92.20 % for slab assembly. In a case study on a real PC construction site, the model demonstrated high recognition accuracies: 100 % for lifting, 85.83–100 % for rigging, and 93.75–100 % for assembly operations. The paper contributes to PC construction management theory by applying computer vision for real-time and automated work recognition and analysis.

1. Introduction

Precast concrete (PC) construction is a building method in which concrete components prefabricated in a factory are assembled on-site [1–3]. Off-Site Construction (OSC) includes not only PC construction, but also modular construction, panelized systems, and prefabricated MEP (Mechanical, Electrical, and Plumbing) systems [4,5]. It gained popularity in the construction industry due to its advantages, such as shortened construction time, improved quality, and cost reduction [6–9]. The installation of PC components is a crucial process in PC construction, consisting of rigging, lifting, assembly, and unrigging installation actions [10,11]. The efficiency and accuracy of these installation actions are vital factors determining project success, making effective monitoring and management of the PC component installation process essential.

However, current monitoring practices for PC component installation face several challenges. According to studies by Mäki and Kerosuo [12] and Styhre [13], construction site managers spend 80 to 90 % of their working hours on office tasks, leaving limited time for on-site monitoring. This makes it difficult to allocate sufficient time for on-site monitoring and raises concerns about the accuracy and consistency of the collected information. Moreover, the lack of real-time, accurate monitoring data hinders effective decision-making and timely interventions in the construction process.

To address these challenges, there is a growing need for automated monitoring systems that can provide real-time, accurate data on PC component installation actions. Such systems could significantly enhance construction management practices by improving progress tracking, workflow optimization, safety monitoring, quality control, and stakeholder communication.

Recent attempts to introduce various technologies for installation action monitoring in construction sites showed limitations in accurately identifying detailed actions in complex processes such as PC component installation. These limitations primarily stem from two factors:

(1) Limited application of computer vision technology in PC construction: Current technologies mainly focus on simple object detection or basic motion recognition, failing to effectively analyze complex installation actions. Studies by Liu et al. [14] and Lee et al. [15] demonstrate that current models are limited to simple object recognition tasks such as detecting worker helmet use or PC column installation. While Wang et al. [16] and Zhang et al. [17] attempted to monitor precast wall installation processes and predict alignment trajectories for PC column

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installation, respectively, they still fall short in understanding the overall flow and context of the installation actions, focusing only on individual component positioning or utilizing information from a single perspective.

(2) Lack of specialized data for PC component installation actions: Existing datasets primarily focus on elements of general construction sites, failing to reflect the specificity of PC component installation actions. Datasets constructed by Duan et al. [9] and Xiao et al. [18] focus on workers, equipment, materials, and heavy machinery in general construction sites. However, these datasets do not include elements specific to PC construction methods, presenting a fundamental limitation in reflecting the uniqueness of PC component installation actions. Meanwhile, PC construction site datasets built by Jang et al. [4] and Yan et al. [19] include PC components and installation tools but fail to sufficiently reflect the continuity of the PC component installation process or the interdependence between installation actions.

These technological limitations and data deficiencies present significant challenges to comprehensively understanding and modeling the complex characteristics of PC component installation actions.

To address these issues, this study aims to construct a comprehensive dataset called PCI (Precast Concrete Installation)-Dataset for automatic recognition of PC component installation actions and to evaluate the performance of Transformer-based video recognition models using this dataset. Specifically, the authors constructed a PCI-Dataset based on actual video data from PC component installation sites and evaluated its performance using VideoMAE V2, TimeSformer, ViViT, MVIT V2, Uniformer V2, and Video SWIN models. Additionally, the authors tested the practicality of the developed model through application in an actual field setting. This study contributes to the enhanced installation action monitoring and management in construction sites and improved productivity by accelerating the digitalization and automation of the construction industry. By providing a foundation for more accurate and detailed monitoring of PC installation actions, this research aims to bridge the gap between academic advancements in computer vision and the practical needs of construction management professionals.

2. Related works

2.1. Deep learning models for OSC installation action recognition

Research on deep learning models for offsite construction (OSC) installation action recognition is divided into factory and on-site environments. Studies in factory environments focus on quality control, productivity improvement, and ergonomic analysis. In contrast, research in on-site environments typically concentrates on safety management, installation progress monitoring, interruption management, and process guidance.

In factory environments, Lee et al. [15] (2020) developed an automatic quality inspection system for detecting cracks and damages in PC components, while Martinez et al. [20] proposed a vision-based system for automatically inspecting screw fastening in lightweight steel frame manufacturing. For productivity improvement and progress monitoring, Zheng et al. [21] developed a Mask R-CNN model to count installed modules in modular integrated structures automatically. Fini et al. [22] presented a model for real-time module tracking. Panahi et al. [23] developed a computer vision-based model for tracking volumetric modules through various stations along the production line. Park et al. [24] proposed a model for recognizing stages of the module assembly process in a virtual factory environment. Martinez et al. [25] developed a model for tracking the progress of floor panel production. In the field of ergonomic analysis, Chu et al. [26] developed a monocular vision-based framework for evaluating worker postures in modular construction factories.

In on-site environments, Liu et al. [25] developed a system using an extended YOLOv3 algorithm to detect workers' PPE usage for safety management in prefabrication construction sites. For installation progress monitoring. For installation progress monitoring, Wang et al. [27] proposed a framework using Faster R-CNN to monitor the installation process of precast walls. In the area of interruption management, Yan and Zhang [28] developed a system using Faster R-CNN to detect and manage various types of interruptions in prefabricated building projects. For process guidance, Zhang et al. [17] proposed a deep learning-based system to predict alignment trajectories during PC column installation.

Existing OSC installation action recognition research primarily uses CNN, LSTM, or combinations of these models, focusing on object detection or recognition of specific tasks, limiting the recognition of installation processes considering the continuity of the entire operation. In particular, there is a lack of research on datasets and model evaluations that can distinguish and recognize continuous installation action stages such as rigging, lifting, assembly, and unrigging in the precast component installation process.

2.2. Datasets for construction site work recognition

Datasets for construction site installation action recognition can be broadly categorized into worker behavior recognition and safety monitoring, construction equipment operation recognition and pose estimation, construction site object detection, and PC and modular construction fields.

As for worker behavior recognition and safety monitoring datasets, Yang et al. [29] developed a dataset including 1595 video clips and seven action classes for identifying unsafe behaviors in construction projects. Luo et al. [30] used about 1000 video clips to build a computer vision-based worker activity assessment model. Bonyani et al. [31] conducted a study to detect unsafe behaviors of construction workers using 2806 images from the SODA dataset and 1595 video clips from the CMA dataset. Wu et al. [32] presented a benchmark dataset for safety helmet wear detection, including 3174 images.

As for construction equipment operation recognition and pose estimation datasets, Ghelmani and Hammad. [33] developed a self-supervised learning model for construction equipment operation recognition using 1036 video clips. Roberts and Golparvar-Fard. [34] built an end-to-end vision-based system for the detection, tracking, and activity analysis of earthmoving equipment using ten videos. Assadzadeh et al. [35] developed an excavator pose estimation model using up to 15,000 synthetic and 824 real images. Luo et al. [36] conducted a study on full-body pose estimation of construction equipment using 1281 images (augmented to 6405). Soltani et al. [37] researched to estimate excavators' skeletons using 30,600 synthetic images. Xiao et al. [18] presented a methodology for automatically detecting construction video highlights by integrating machine tracking and CNN feature extraction.

As for construction site object detection datasets, Duan et al. [9] developed the SODA dataset, including 19,846 images, for object detection in construction sites. Yan et al. [19] presented the Construction Instance Segmentation (CIS) dataset of 50,000 images. Tajeen and Zhu. [38] developed a dataset using 2000 images to measure construction equipment recognition performance. Fang et al. [39] built an automatic detection model for workers and heavy equipment using over 10,000 image frames. Luo et al. [40] conducted a study to recognize various construction activities through relevance networks of construction-related objects detected by CNN using 7790 images. Zhu et al. [41] developed an integrated detection and tracking system for workers and equipment in construction site videos using 44,597 image frames.

As for PC and modular construction datasets, Jang et al. [6] developed a PC project image dataset including 12,000 high-resolution images. Baek et al. [42] researched module installation productivity monitoring in OSC using about 28 h of video. Lee et al. [43] developed a PC component crack detection dataset of 4000 images. Panahi et al. [23]

researched to identify modular construction worker tasks using computer vision. Park and Ergan [44] presented an approach to generate training data for scene understanding in modular construction using 84,000 synthetic images. Panahi et al. [45] conducted a study to detect bottlenecks in modular construction factories through computer vision using 420 h of surveillance video. Chen et al. [46] built a real-time process monitoring and problem feedback system in OSC using 3200 images.

Despite the progress, there remains a significant gap in the field of OSC construction datasets. Specifically, there is a lack of comprehensive datasets that cover the entire process of precast component installation actions in OSC sites. The dataset by Baek et al. [42] is a step forward as it focuses on module installation, but it has limitations in not addressing the entire installation process in detail, such as rigging, lifting, assembly, and unrigging, which are specific to PC.

An ideal OSC dataset should have the following characteristics: long-term continuous recording, inclusion of various installation action stages (rigging, lifting, assembly, unrigging), multi-angle recording, reflection of various environmental conditions, capture of interactions between workers and equipment, and high-resolution video. Table 1 summarizes existing datasets for behavior or object recognition for use in OSC projects.

2.3. Summary

Based on our review of existing literature, several key gaps have been identified in the research on Off-Site Construction (OSC) installation task recognition: (1) the scope of continuous installation processes, (2) lack of datasets encompassing the entire process, (3) limited model evaluation for recognizing continuous installation task stages, and (4) insufficient consideration of environmental and contextual factors. To address these gaps, our study aims to: (a) develop an OSC installation process dataset that includes task stages such as rigging, lifting, assembly, and unrigging, (b) propose and evaluate deep learning models for recognizing continuous installation task stages, (c) capture OSC environments and worker-equipment interactions using various high-resolution videos, and (d) provide a dataset and model evaluation framework for future research and applications. Through these efforts, our study seeks to contribute to the understanding and improvement of OSC processes.

3. Methods

This study was conducted in three main stages, as shown in Fig. 1. First, the authors constructed the PCI-Dataset, a video dataset for recognizing PC component installation actions. Second, the authors evaluated the performance of deep learning models for automatic recognition of PC component installation actions using the constructed dataset. Finally, the authors tested the effectiveness of the selected best-performing model by applying it to an actual PC component installation site.

3.1. PCI-dataset construction

This study constructed the PCI-Dataset, a dataset for automatically recognizing PC component installation actions. The dataset construction process proceeded in the following steps. Through a literature review [10,11,47,48] and interviews with field experts, the authors defined the main installation actions of PC component installation and established a classification system. The installation process was divided into four action stages (ring clutch rigging, lifting, assembly, unrigging) and combined with three types of components (column, beam, slab) to define a total of 12 installation actions. Clear definitions and descriptions for each installation action are presented in Table 2.

3.1.1. Collection of PC component installation videos

To construct the image dataset, a systematic data collection process from various sources was necessary [49]. Following this process, the authors conducted filming for two months at one domestic construction site applying the PC method. One thousand-seven videos (about 700 h) were collected using four action cameras and one smartphone. The collected videos were classified by PC component type, consisting of 362 beams, 207 slabs, and 438 columns of videos.

3.1.2. Preprocessing of PC component installation actions

A systematic preprocessing was performed to transform the collected raw video data into a form suitable for deep learning model training. This process was based on the methodologies of Jang et al. [6], Xiao et al. [18], and Duan et al. [9], with modifications and applications. The preprocessing consisted of two stages: manual annotation work and data structuring and video clip segmentation.

- Manual annotation work and data structuring: The research team analyzed all videos over four weeks to identify each installation action's start and end points and assign labels. Clear start and end criteria, as presented in Table 3, were applied to ensure consistency of annotations. The annotation data was organized into an Excel file, including information such as folder name, file name, component type, installation action type, and start and end times (Table 4).

- Video clip segmentation: Using a Python script developed based on the Excel data, the original videos were automatically segmented into individual clips of 5 s in length. This duration was chosen based on our field data analysis of PC installation actions (ranging from ~2 to 5 s) and consideration of clip lengths in widely used datasets (e.g., UCF101, Kinetics, HMDB51: 3–10 s). This approach aimed to

Table 1
Existing datasets for behavior or object recognition in OSC construction.

Ref.	Recognition Type	Object	Number of Clips	Algorithms	Metrics
Jang et al. [6]	Object Detection	PC Component	12,000 images	YOLOv5, YOLOX, Faster R-CNN, Double Head R-CNN	mAP(97 %), Robustness(96 %), fps(23)
Baek et al. [42]	Object Detection, Behavior Recognition	Module, Worker	28 h 29 min 27 s video	YOLO v4	Precision(91 %), Recall(91 %), F1-score(91 %), mAP(93 %)
Lee et al. [44]	Object Detection	PC components	4000 images	SSD	Recall(78 %), Precision(75 %)
Panahi et al. [23]	Behavior Recognition	Modular construction workers	–	CNN, SSD	Accuracy(90 %)
Park & Ergan. [44]	Object Detection	Modular construction components	84,000 images	CNN-based 3D reconstruction network	Accuracy(97 %)
Panahi et al. [45]	Object Detection	Modular construction workplace	420 h video	SIFT, Mask R-CNN	Workplace occupancy/state(88 %) identification accuracy(91 %), F1-score(89 %)
Chen et al. [46]	Object Detection, Behavior Recognition	Workers, equipment, etc.	3200 images	YOLO V8, ByteTrack	Precision(97 %), Recall(98 %), IDSW(2.5), MOTA(97 %), MOTP(86 %)

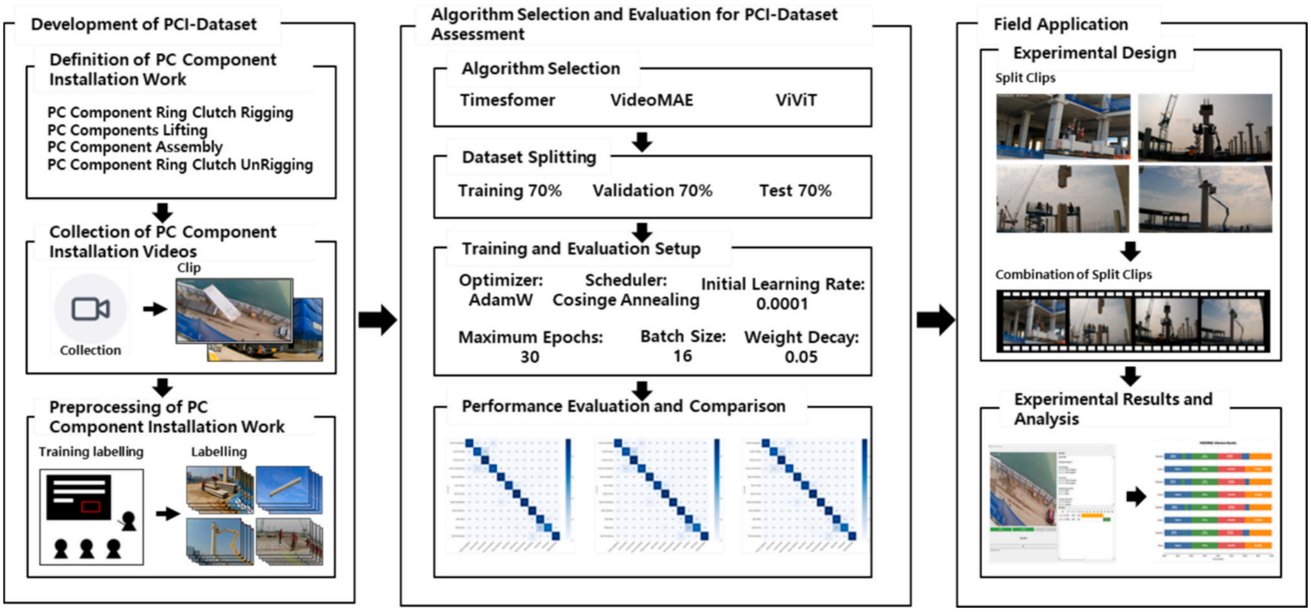

Fig. 1. Development and application process of PCI-Dataset.

Table 2
Definition of installation actions.

Activity	Description
Column -Rigging (C-R)	Connecting rigging equipment to a column for lifting
Column-Lifting (C-L)	Lifting a column using a crane with rigging equipment
Column-Assembly (C-A)	Installing and adjusting a column at the designated location
Column-Unrigging (C-UR)	Removing rigging equipment from a column after installation
Beam-Rigging (B-R)	Connecting rigging equipment to a beam for lifting
Beam-Lifting (B-L)	Lifting a beam using a crane with rigging equipment
Beam-Assembly (B-A)	Installing and adjusting a beam at the designated location
Beam-Unrigging (B-UR)	Removing rigging equipment from a beam after installation
Slab-Rigging (S-R)	Connecting rigging equipment to a slab for lifting
Slab-Lifting (S-L)	Lifting a slab using a crane with rigging equipment
Slab-Assembly (S-A)	Installing and adjusting a slab at the designated location
Slab-Unrigging (S-UR)	Removing rigging equipment from a slab after installation

Table 3
PC component installation work definition.

Activity	Start Time	End Time
Rigging	The moment the ring clutch appears on the screen	The moment the ring clutch is completely fastened to the component
Lifting	The moment the component starts to lift	Just before the worker grabs the installation control rope
Assembly	The moment the worker grabs the component with their hand or grabs the hoisting rope	The moment the component is fixed in place
Unrigging	Immediately after the component is fixed	The moment the hoisting rope is completely removed from the component

effectively capture varying action durations while maintaining dataset compatibility. This approach ensured optimal capture of actions with varying durations. Irrelevant sections were removed, and all clips were standardized to 1080 × 1080 resolution, 30fps [29]. Through this process, a total dataset consisting of 56,901 clips was generated.

3.1.3. Final dataset composition

Clips were selected to include 12 detailed installation actions (three component types by four action stages) in a balanced manner. Considering the limitations of computer performance and computational resources, the final dataset size was set to 12,791. This dataset clearly defines each installation action's start and end points, providing a reliable standard for deep learning model training and evaluation.

3.2. Algorithm selection and evaluation for PCI-dataset assessment

This study used video analysis models including VideoMAE, ViViT, TimeSformer, MVIT V2, UniFormer V2, and Video Swin V2 to evaluate the effectiveness of PCI-Dataset and analyze the performance of PC component installation action recognition. While traditional approaches such as 3D-CNN and LSTM have been widely used in construction activity recognition, they have limitations in modeling long-range dependencies and handling long sequences efficiently [27]. These Transformer-based models were chosen based on their established performance in video understanding tasks and their ability to address these limitations. However, our primary focus is on assessing the quality and usefulness of PCI-Dataset, rather than comparing algorithm performances. Through these models, we evaluated the characteristics and potential applications of PCI-Dataset from various angles. Additionally, we included performance results of CNN-based models (I3D, C3D, R(2 + 1)D), hybrid models (CNN + LSTM, TSM), as well as traditional 3D-CNN and LSTM models in our evaluation for a comprehensive comparison and to provide a broader perspective on how different architectural approaches interact with our dataset (see Section 4.2).

3.2.1. Algorithm selection

VideoMAE V2 is a pre-training model that utilizes self-supervised learning. It randomly masks some patches along the time and spatial axes in the input video and learns general features of the video through the process of reconstructing the masked patches through an encoder-decoder structure. High performance can be expected by fine-tuning the pre-trained encoder for the task of installation action recognition [50]. This model was chosen because it can effectively learn from a large amount of unlabeled video data, which is expected to understand various situations at PC component installation sites well.

TimeSformer is a Transformer-based video recognition model that explicitly models long-term temporal dependencies through temporal

Table 4
PC component installation video labeling summary.

Date	Title	File name	Component Type	Activity	Start	End
20,220,606	Project1	2.3GH030221	Column	Assembly	0:00	10:24
20,220,606	Project1	2.4GH040221	Beam	Lifting	0:00	10:35
20,220,606	Project1	2.5GH010222	Slab	Assembly	0:00	7:03
20,220,606	Project1	2.6GH010222	Slab	Lifting	0:00	7:03
20,220,606	Project2	2.7GH010222	Slab	Assembly	0:00	6:00
20,220,606	Project2	2.8GH040222	Slab	Rigging	0:00	6:00
20,220,606	Project2	2.9GH040222	Slab	Unrigging	0:00	9:10

attention. It divides the input video into spatiotemporal patches and then sequentially applies spatial and temporal attention to each patch for encoding. This enhances the model's expressiveness by separating spatial axis modeling, which extracts scene features, and temporal axis modeling, which captures changes over time [51]. This model was selected because it is expected to capture the sequential characteristics and long-term dependencies of PC component installation actions well.

ViViT is a video recognition model based on the Transformer structure. It divides each frame into patch units and encodes them through a Transformer encoder to extract features of the entire video. It learns the relationships between patches through a multi-head attention mechanism and outputs the final recognition result through MLP layers [52]. This model was selected because it is expected to effectively learn the complex spatiotemporal relationships of PC component installation actions.

MViT v2 is a video recognition model that extracts features from various spatiotemporal scales through a hierarchical structure. It uses a pooling attention mechanism to increase computational efficiency while capturing extensive spatiotemporal relationships. Additionally, it improves efficiency by independently processing spatial and temporal dimensions through separated spatiotemporal attention [53]. This model was chosen because it can simultaneously capture various detailed actions and the overall flow of the PC component installation process.

Uniformer v2 is a unified architecture for image and video processing. It employs a mixed attention mechanism that combines local and global attention, and effectively encodes temporal position information through dynamic position encoding. It also possesses multimodal learning capabilities that can simultaneously process image and video data [54]. This model was selected because it can simultaneously understand the detailed actions and overall process of PC component installation, and accurately capture the installation sequence and changes over time.

Video Swin is a video transformer model with a hierarchical structure. It uses a shifted window technique to maintain connectivity between windows while using local windows for efficient computation. It processes spatial and temporal information simultaneously through 3D spatiotemporal attention and effectively encodes position information using relative position bias [55]. This model was chosen because it can capture both the local characteristics and overall context of PC component installation, and can simultaneously learn the temporal and spatial relationships of the installation process.

3.2.2. Dataset split

PCI-Dataset was split into training (70 %), validation (10 %), and evaluation (20 %) sets [56,57]. Stratified sampling was applied to ensure that each subset evenly includes 12 installation action classes. This ensures balanced learning and evaluation of the model.

3.2.3. Training and evaluation settings

Model implementation and training were conducted using the PyTorch framework. The main training parameters are as follows. AdamW was used as the optimizer, and Cosine Annealing was applied as the learning rate scheduler. The batch size was set to 16, the initial learning rate to 0.0001, the maximum number of epochs to 30, and the weight decay to 0.05 [58]. These parameter settings ensure that the model learns stably, prevents overfitting, and performs optimally.

3.2.4. PCI-dataset evaluation method

This study evaluated the prediction performance of the three models implemented using PCI-Dataset. Standard performance indicators such as accuracy, precision, recall, and F1 score were used for evaluation. Eqs. (1), (2), (3), and (4) were used to calculate each indicator. The confusion matrix was used to analyze the prediction performance and error patterns for each class in detail. In addition, to evaluate the possibility of actual field application, the authors collected separate test data from a PC construction project site and analyzed the performance of each model comprehensively.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{Prcision \times Recall}{Prcision + Recall} \tag{4}$$

4. Results

4.1. PCI-dataset profile

As explained in the previous section, the PCI-Dataset comprises 12,791 video clips. Each installation action class accounts for 8.1 % to 8.5 % of the entire dataset, showing a well-balanced distribution. By PC component, it shows an even distribution of about 33 %, and by the installation action stage, about 25 %. This composition comprehensively reflects various PC component installation action situations (Table 5). The total duration of all videos in the dataset is 1065.55 min, with each activity category having approximately equal representation in terms of video count and duration (Table 6).

This dataset was designed to reflect the variability and diversity of actual installation environments, including various weather conditions, time zones, workers, and equipment. It was also designed to contribute to improving the generalization ability and robustness of the six models,

Table 5
Profile of the PCI-Dataset.

Installation activities	Beam	Slab	Column	Total
Rigging	1071(8.37 %)	1049(8.20 %)	1068(8.35 %)	3188(24.92 %)
Lifting	1070(8.36 %)	1057(8.26 %)	1076(8.41 %)	3203(25.04 %)
Assembly	1081(8.45 %)	1072(8.38 %)	1081(8.45 %)	3234(25.28 %)
Unrigging	1065(8.33 %)	1038(8.12 %)	1063(8.31 %)	3166(24.75 %)
Total	4287(33.52 %)	4216(32.96 %)	4288(33.52 %)	12,791(100 %)

Table 6
Analysis of work activities and video duration by categories.

Categories	Activity	Total time(min)
Beam	Rigging	89.15
	Lifting	85.10
	Assembly	90.05
	Unrigging	88.45
Slab	Rigging	87.25
	Lifting	88.05
	Assembly	89.20
	Unrigging	86.30
Column	Rigging	89.00
	Lifting	89.40
	Assembly	90.05
	Unrigging	88.35
Total		1065.55

Note: All videos were shot at a resolution of 1080 × 1080 and a frame rate of 30 fps.

i.e., VideoMAE V2, TimeSformer, ViViT, MVIT V2, Uniformer V2, Video Swin V2. Fig. 2 shows representative frames of 12 installation actions.

4.2. Performance of PC component installation action recognition based on PCI-dataset

4.2.1. Overall performance comparison

As shown in Table 7, VideoMAE V2 showed the best performance with an accuracy of 98.10 %. Other Transformer-based models also performed well, with UniformerV2, Video SWIN, and MVIT V2 achieving accuracies of 94.20 %, 93.25 %, and 92.81 %, respectively. The excellent performance of VideoMAE V2 is analyzed due to effective pre-training through masked autoencoding and the ability to model long-term dependencies through the Transformer structure. The proposed Transformer-based methods all achieved high recognition accuracy of over 87 %, outperforming traditional approaches such as I3D (79.75 %), R(2 + 1)D (80.76 %), and CNN + LSTM (87.34 %). This result demonstrates that the proposed PCI-Dataset and Transformer-based video recognition models can be very effectively applied to PC component installation action recognition, showing significant improvements over conventional methods.

4.2.2. Performance in detail by installation actions and models

Tables 8, 9, 10 shows the recognition performance of each model for 12 detailed installation actions. The models demonstrated varying performance across different actions.

VideoMAE V2 (1) showed high performance across most actions, with accuracies often above 95 %. TimeSformer (2) and ViViT (3) performed well in specific actions. TimeSformer achieved 92.3 % accuracy in Column Assembly and 92.9 % in Beam Rigging, while ViViT reached 95.8 % in both Beam Rigging and Slab Assembly.

UniformerV2 (4), Video SWIN (5), and MVIT V2 (6) also showed competitive performance. UniformerV2 achieved 98.75 % accuracy in Column Assembly, Video SWIN 98.80 % in Beam Unrigging, and MVIT V2 99.39 % in Slab Unrigging.

These six Transformer-based models generally outperformed traditional approaches like CNN + LSTM (7), I3D (8), C3D (9), R(2 + 1)D (10), and TSM (11). However, some traditional models performed well in certain actions, such as I3D achieving 98.47 % accuracy in Column Rigging.

The models showed lower performance for Column Lifting and Slab Rigging actions. ViViT achieved 79.0 % accuracy in Column Lifting and 70.1 % in Slab Rigging. This might be due to these actions having less distinct visual cues or higher similarity to other actions.

In contrast, actions like Beam Assembly, Beam Unrigging, and Slab Unrigging saw higher performance across most models. VideoMAE V2 achieved 98.1 %, 98.2 %, and 98.7 % accuracy in these actions respectively.

These performance differences may be attributed to the structural characteristics and learning methods of each model. The performance of Transformer-based models in this task indicates the potential of these architectures in recognizing complex actions in construction scenarios. The performance of some traditional models in certain actions suggests that combining different approaches might be beneficial for comprehensive action recognition in PC installation.

4.2.3. Confusion matrix analysis

Figs. 3, 4, and 5 represent the confusion matrices of three models. These confusion matrices imply that most errors stemmed from confusion between similar installation actions. Confusion between rigging and unrigging actions and between lifting and assembly actions was notably observed because these pairs share visually similar characteristics. These results suggest the need to define the boundaries between installation actions more clearly and reflect this in annotation and learning strategies.

4.2.4. Demonstration of PC installation action recognition using PCI-dataset

Fig. 6 shows the recognition results of the VideoMAE V2 model for representative installation actions among the 12 main actions of PCI-Dataset. Each row represents the same installation action at different times, and the “Prob” indicated on the right represents the model’s prediction confidence. The VideoMAE V2 model trained using PCI-Dataset performed accurate recognition with a high probability of over 90 % for all installation actions. Particularly noteworthy points are as follows:

- It recognized all installation actions related to various PC components included in the dataset, such as columns, beams, and slabs, with high accuracy. This suggests that the PCI dataset reflects the diversity of PC components well.
- It accurately distinguished various installation action stages occurring during the installation process of each component, such as ring clutch rigging, lifting, assembly, and unrigging. This shows that the PCI-Dataset comprehensively covers the entire process of PC installation actions.
- It effectively captured and utilized for recognition of the characteristic visual elements of each installation action included in the PCI-Dataset, such as crane use, worker movements, and changes in component position. This means that the dataset captures various visual elements of the site well.
- It accurately identified subtle differences between similar installation actions, clearly distinguishing, for example, column lifting from beam lifting or slab assembly from beam assembly. This indicates that PCI-Dataset provides detailed information that can capture subtle differences between similar installation actions.

These results suggest that the PCI-Dataset effectively represents various installation actions in the PC component installation process. Based on this, the VideoMAE V2 model can very accurately identify and recognize installation actions occurring in various construction environments and action stages.

4.3. Field applicability of the PCI-dataset

To evaluate the actual field applicability of PCI-Dataset developed in this study and verify the versatility of the dataset, the authors conducted a case study on a domestic PC construction project site.

4.3.1. Experimental design

This experiment was conducted on a new construction site of a PC structure logistics center with a total floor area of 264,000 m^2 and five stories above ground. The actual field applicability of PCI-Dataset with the VideoMAE V2 model was evaluated in this experiment. To do so, the

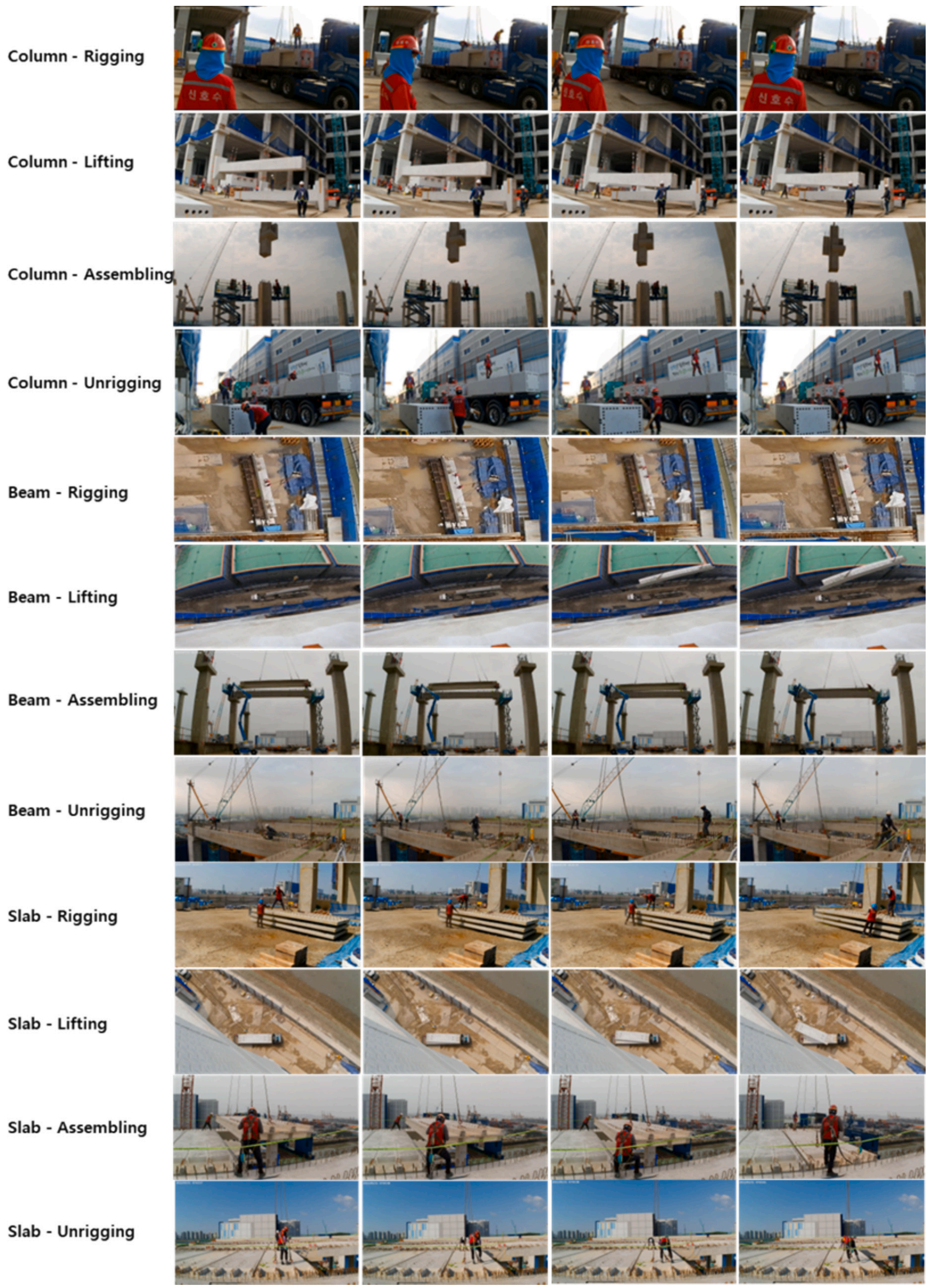

Fig. 2. Representative frame sequences of PC component installation activities in the PCI-Dataset.

Table 7
Performance comparison on the PC component installation dataset.

No	Model	Accuracy	Precision	Recall	F1 Score
1	Videomae V2	98.10 %	88.23 %	88.92 %	88.96 %
2	Uniformer V2	94.20 %	94.31 %	94.20 %	94.19 %
3	Video SWIN	93.25 %	93.43 %	93.25 %	93.19 %
4	MVIT V2	92.10 %	92.36 %	92.10 %	91.98 %
5	TimeSformer	87.76 %	88.20 %	87.72 %	87.96 %
6	ViViT	88.45 %	88.97 %	88.45 %	88.71 %
7	C3D	90.50 %	90.70 %	90.50 %	90.41 %
8	CNN + LSTM	87.34 %	87.58 %	87.34 %	87.15 %
9	I3D	79.75 %	80.23 %	79.77 %	79.61 %
10	TSM	81.07 %	82.28 %	81.07 %	80.86 %
11	R(2 + 1)	80.76 %	81.15 %	80.76 %	80.35 %

authors first recorded PC component installation actions through a 7-day on-site observation. During this period, beam (n = 48), slab (n = 52), and column (n = 42) installations were observed. The collected videos were classified by component type and installation action stage reflecting the structure of PCI-Dataset and then reconstructed into four consecutive installation action sequences in the order of rigging-lifting-assembly-unrigging. Additionally, a video similarity analysis was performed to quantitatively compare the case study videos with the PCI-Dataset. To facilitate comprehensive process analysis, A Value Stream Map (VSM) was developed for each component type, incorporating detailed time statistics including quartiles and 95 % confidence intervals.

Also, the authors developed an installation action recognition prototype system (Fig. 7) based on the VideoMAE V2 model trained with PCI-Dataset. This prototype was used to evaluate the dataset's generalization ability and field applicability, with the model's recognition accuracy and consistency as the main evaluation criteria.

4.3.2. Experiment results

The VideoMAE V2 model trained with PCI-Dataset showed high recognition accuracy at the actual PC component installation site. As shown in Fig. 8, which analyzes a continuous sequence of four installation action processes over 8 min, the lifting action showed 100 % accuracy, rigging action 85.83 %–100 %, assembly action 93.75 %–100 %, and unrigging action 91.67 % accuracy. This suggests that the PCI-Dataset effectively captures the characteristics of each installation action stage and appropriately reflects the diversity and complexity of the actual site. As a result, it was proven that the trained model can perform stably in actual sites, even with continuous and sequential activities. Notably, a video similarity analysis between the case study videos and the PCI-Dataset revealed a low average similarity score of 0.43, further emphasizing the model's ability to generalize to significantly different real-world scenarios. To provide a more comprehensive evaluation, we extended our analysis to the complete dataset collected over a 7-day period. The results of this extended analysis are presented in Table 11.

This extended analysis confirms the model's high performance over a longer period, maintaining an average of 98.1 % recognition accuracy across all activities. The data also provides valuable insights for construction practitioners, supporting data-driven decision-making in

Table 8
Performance comparison of various models for column-related installation actions.*

Activity	Metric	1	2	3	4	5	6	7	8	9	10	11
C-R	Acc	97.7	88.8	96.4	95.5	96.4	94.0	89.2	98.4	79.0	68.4	88.5
	Pre	83.7	84.9	80.1	89.8	87.9	88.2	74.1	77.7	70.9	76.6	55.6
	Rec	89.7	88.7	96.4	92.5	96.4	94.0	89.2	86.8	79.0	72.3	68.3
	F1	86.6	86.8	87.5	89.8	92.0	91.0	80.9	77.7	74.7	76.6	55.6
C-L	Acc	97.0	83.8	79.0	88.3	78.4	79.0	60.8	96.9	59.8	66.9	62.1
	Pre	82.9	80.0	84.0	91.0	91.6	92.9	77.6	96.9	68.9	50.9	81.4
	Rec	81.1	83.7	79.0	89.6	78.4	79.0	80.8	96.9	59.8	57.8	70.4
	F1	82.2	81.8	81.4	91.0	84.5	85.4	68.2	96.9	64.1	50.9	81.4
C-I	Acc	98.4	92.3	89.8	98.7	95.1	95.7	97.5	82.4	91.5	92.9	92.4
	Pre	91.2	94.7	93.7	95.1	98.1	95.7	93.6	73.0	92.6	94.5	96.3
	Rec	90.3	92.3	89.7	96.9	95.1	95.7	97.5	77.4	91.5	93.7	94.4
	F1	90.7	93.5	91.6	95.1	96.6	95.7	95.5	73.0	92.1	94.5	96.3
C-UR	Acc	97.6	83.7	80.2	88.2	94.6	94.6	82.0	88.4	76.0	80.2	83.0
	Pre	87.3	89.4	89.3	94.0	87.2	86.8	83.5	86.8	73.4	85.0	85.0
	Rec	82.9	83.7	80.2	91.0	94.6	94.6	82.0	87.6	76.0	82.5	84.0
	F1	85.0	86.4	84.5	94.0	90.8	90.5	82.7	86.8	74.7	85.0	85.0

* C-R: Column Rigging C-L: Column Lifting C-I: Column Assembly C-UR: Column Unrigging B-R: Beam Rigging B-L: Beam Lifting B-I: Beam Assembly B-UR: Beam Unrigging S-R: Slab Rigging S-L: Slab Lifting S-I: Slab Assembly S-UR: Slab Unrigging.

Table 9
Performance comparison of various models for beam-related installation actions.

Activity	Metric	1	2	3	4	5	6	7	8	9	10	11
B-R	Acc	98.4	92.9	95.8	93.7	98.2	99.4	96.4	83.2	59.2	78.2	81.8
	Pre	87.7	91.5	89.8	99.4	93.1	87.8	88.9	92.2	81.1	92.8	64.6
	Rec	93.9	92.9	95.8	96.5	98.2	99.4	96.4	87.5	59.2	84.9	72.2
	F1	90.7	92.2	92.7	99.4	95.6	93.2	92.5	92.2	68.5	92.8	64.6
B-L	Acc	97.1	85.4	86.7	97.9	85.5	76.5	79.5	91.7	80.7	92.1	68.0
	Pre	89.4	88.4	91.7	87.9	98.6	96.2	92.9	93.4	63.8	63.2	78.3
	Rec	81.9	85.4	86.7	92.7	85.5	76.5	79.5	92.5	80.7	75.0	72.8
	F1	85.5	86.9	89.1	87.9	91.6	85.2	85.7	93.4	71.2	63.2	78.3
B-I	Acc	98.1	93.8	92.8	94.6	94.6	91.6	88.0	93.4	91.0	86.2	78.3
	Pre	88.4	89.0	91.1	94.6	94.6	91.0	88.0	93.4	84.4	82.6	93.4
	Rec	89.4	93.7	92.8	94.6	94.6	91.6	88.0	93.4	91.0	84.4	85.2
	F1	88.9	91.3	91.9	94.6	94.6	91.3	88.0	93.4	87.6	82.6	93.4
B-UR	Acc	98.2	91.1	94.0	94.8	98.8	97.6	92.2	86.1	77.2	75.1	93.3
	Pre	90.6	93.7	94.0	98.8	92.1	90.6	88.5	89.2	92.8	92.2	75.4
	Rec	87.2	91.0	94.0	96.7	98.8	97.6	92.2	87.6	77.2	82.8	83.4
	F1	88.9	92.3	94.0	98.8	95.3	93.6	90.3	89.2	84.3	92.2	75.4

Table 10
Performance comparison of various models for slab-related installation actions.

Activity	Metric	1	2	3	4	5	6	7	8	9	10	11
S-R	Acc	98.7	71.2	70.1	96.9	97.6	98.8	94.0	90.9	89.2	86.7	87.5
	Pre	87.8	91.3	92.8	95.8	93.1	95.9	94.5	96.4	85.1	94.0	92.2
	Rec	97.1	71.2	70.0	96.3	97.6	98.8	94.0	93.6	89.2	90.2	89.8
	F1	92.2	80.4	79.8	95.8	95.3	97.3	94.2	96.4	87.1	94.0	92.2
S-L	Acc	98.1	85.8	88.6	94.4	89.2	85.6	87.9	96.9	71.2	88.5	83.6
	Pre	89.8	71.2	74.0	91.6	96.7	97.9	88.4	94.5	79.	74.2	76.6
	Rec	86.2	85.8	88.6	93.0	89.2	85.6	87.9	95.7	71.2	80.7	80.0
	F1	88.0	77.8	80.6	91.6	92.8	91.3	88.2	94.5	75.3	74.2	76.6
S-I	Acc	98.1	90.9	95.8	94.5	92.2	92.8	81.4	92.2	92.2	80.2	89.4
	Pre	89.6	89.9	89.3	92.8	95.0	93.9	96.4	92.2	81.9	70.6	76.0
	Rec	87.4	90.9	95.8	93.6	92.2	92.8	81.4	92.2	92.2	75.1	82.2
	F1	88.5	90.4	92.4	92.8	93.6	93.3	88.3	92.2	86.7	70.6	76.0
S-UR	Acc	98.7	93.0	92.1	93.7	98.1	99.3	98.7	87.4	89.7	77.9	78.9
	Pre	91.6	93.9	97.4	99.3	92.5	91.6	84.0	100	87.5	92.1	97.5
	Rec	92.5	93.0	92.1	96.4	98.1	99.3	98.7	93.3	89.7	84.4	87.2
	F1	92.0	93.4	94.7	99.3	95.2	95.3	90.8	100	88.6	92.1	97.5

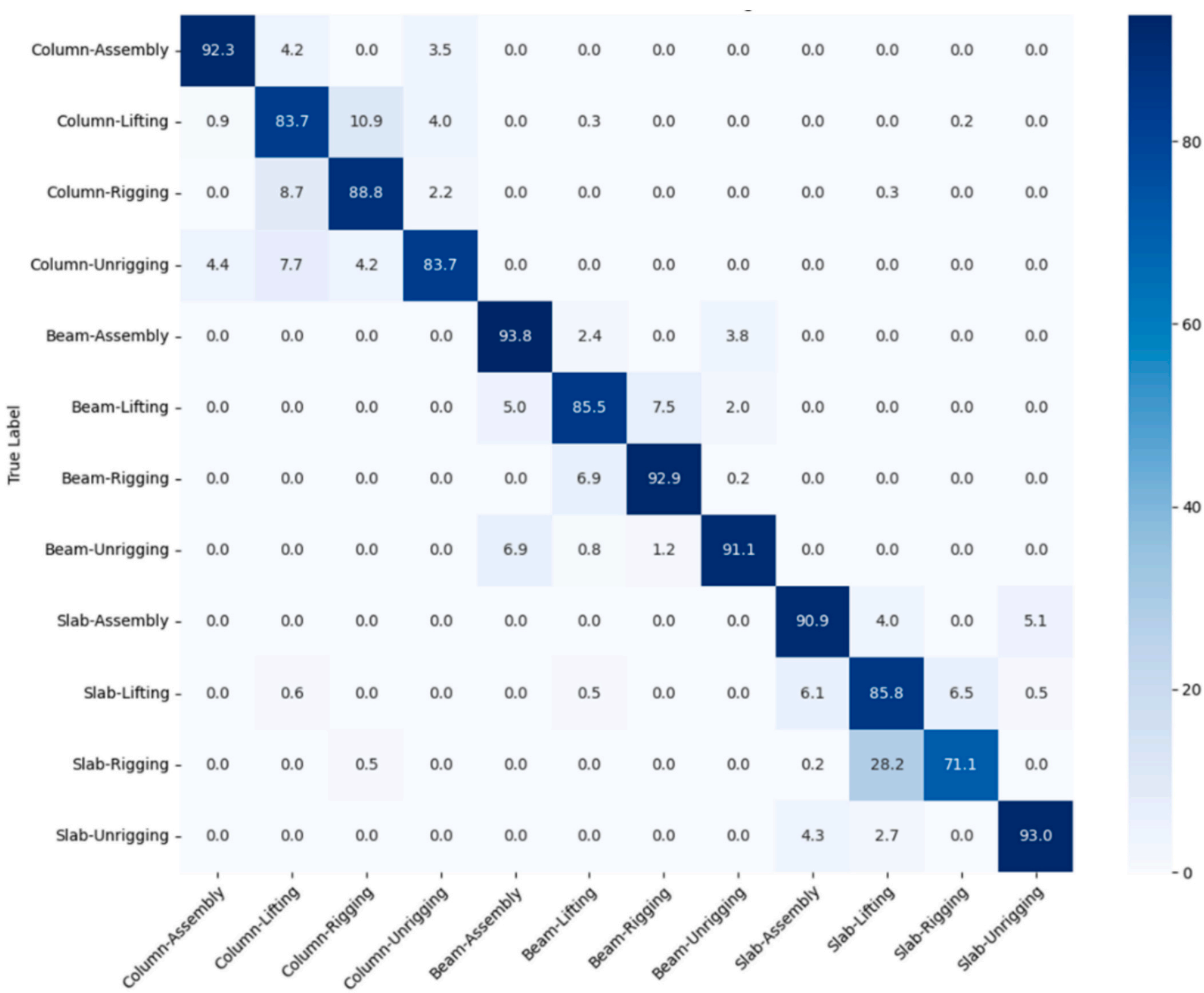

Fig. 3. Timesformer confusion matrix.

several key areas:

- **Productivity improvement:** The analysis reveals significant variations in productivity across different activities. Unrigging operations show the highest productivity, particularly for slabs (43.6 units/h) and beams (8.2 units/h), while assembly operations consistently show lower productivity (1.1 to 4.5 units/h). This suggests potential for targeted improvements in the assembly process.

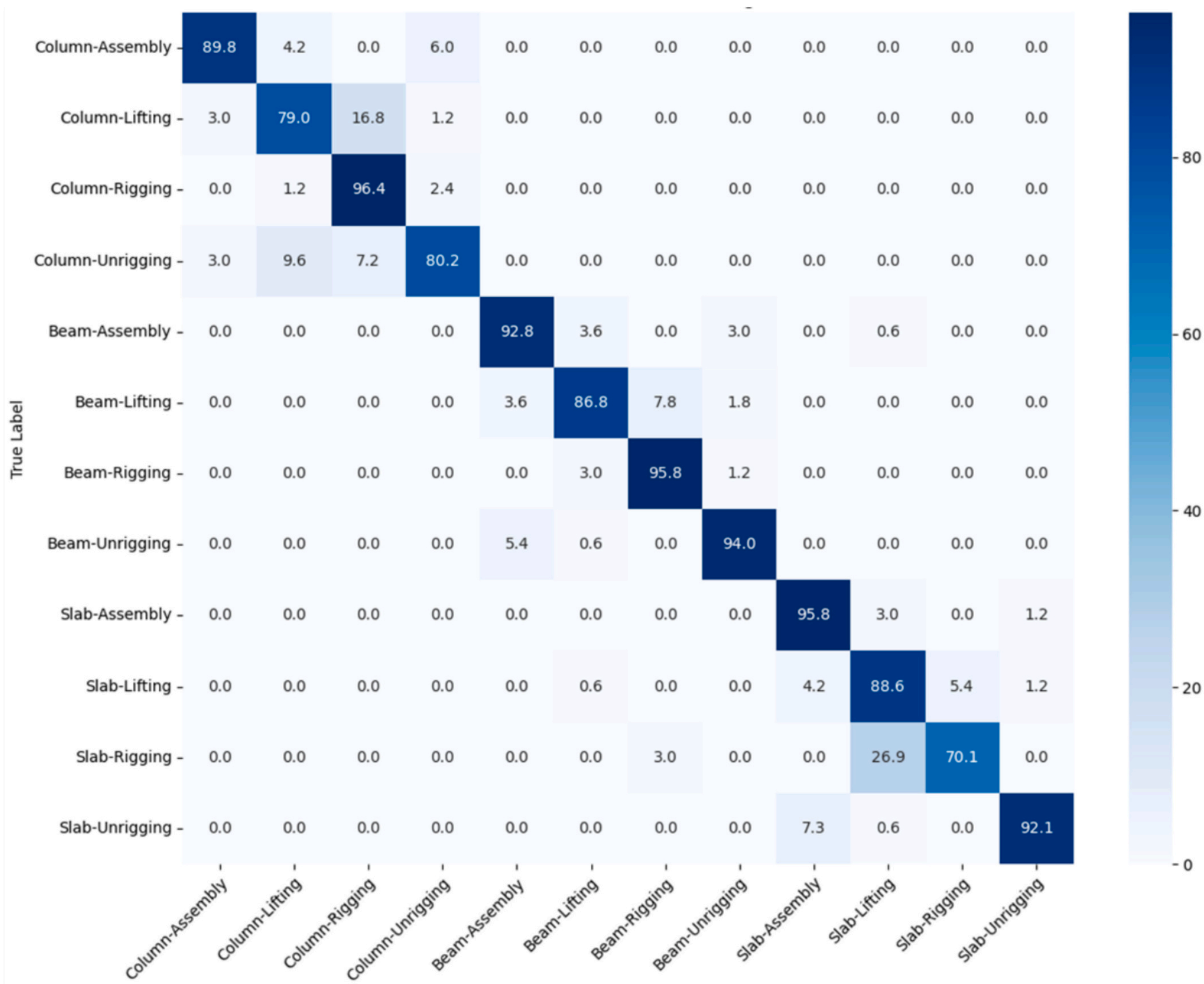

Fig. 4. ViViT confusion matrix.

- **Progress monitoring:** The distribution of work time varies significantly across installation phases. Column assembly takes the longest time (51.3 h), followed by column lifting (39.1 h) and beam assembly (33h). This information enables more accurate real-time progress tracking and scheduling.
- **Workflow optimization:** Assembly operations have the longest average waiting times, with column assembly (442.2 s) and beam assembly (415.8 s) showing the highest values. This indicates potential bottlenecks in the installation process that could be addressed to improve overall efficiency.
- **Component-specific insights:** Slab installation appears to be the most efficient process overall, with high productivity rates and lower waiting times compared to beam and column installation. This information can guide resource allocation and process refinement efforts.

To visualize these patterns in detail, Fig. 9 presents VSM for the beam, slab, and column installation processes, respectively, showing the complete workflow sequence (Loading Bay & Rigging → Waiting → Lifting → Waiting → Assembly → Waiting → Unrigging → Back to Loading Bay) along with corresponding time distributions (min, 25th percentile, median, 75th percentile, max) and waiting periods. For beam installations, assembly operations showed the highest variability (36.0–44.0 min) with a median of 40.0 min, while the total installation time averaged 86.6 min (95 % CI [82.3, 90.9]). Slab installations demonstrated higher efficiency, averaging 36.1 min (95 % CI [34.2, 38.0]), whereas column installations required the longest time, averaging 125.5 min (95 % CI [120.8, 130.2]). These temporal insights, derived from our computer vision-based monitoring system, can support the integration of digital technologies for process improvement, particularly in predicting and preventing excessive waiting periods through real-time monitoring and automated scheduling.

However, notable errors were observed in some installation action processes (Fig. 10). There were cases where the rigging stage was misrecognized between assembly and unrigging actions, and unrigging action was misrecognized in the rigging stage. The main reason would be that the rigging and unrigging actions share visually similar characteristics. These results suggest the need to distinguish between such similar installation actions in PCI-Dataset more clearly.

Overall, these experimental results show that the PCI-Dataset effectively represents the key stages of PC component installation, and models trained on it can perform excellently in real-world settings. The high recognition accuracy (98.1 % overall) combined with detailed temporal insights proves its potential to enhance construction management practices through data-driven decision-making and digital technology integration.

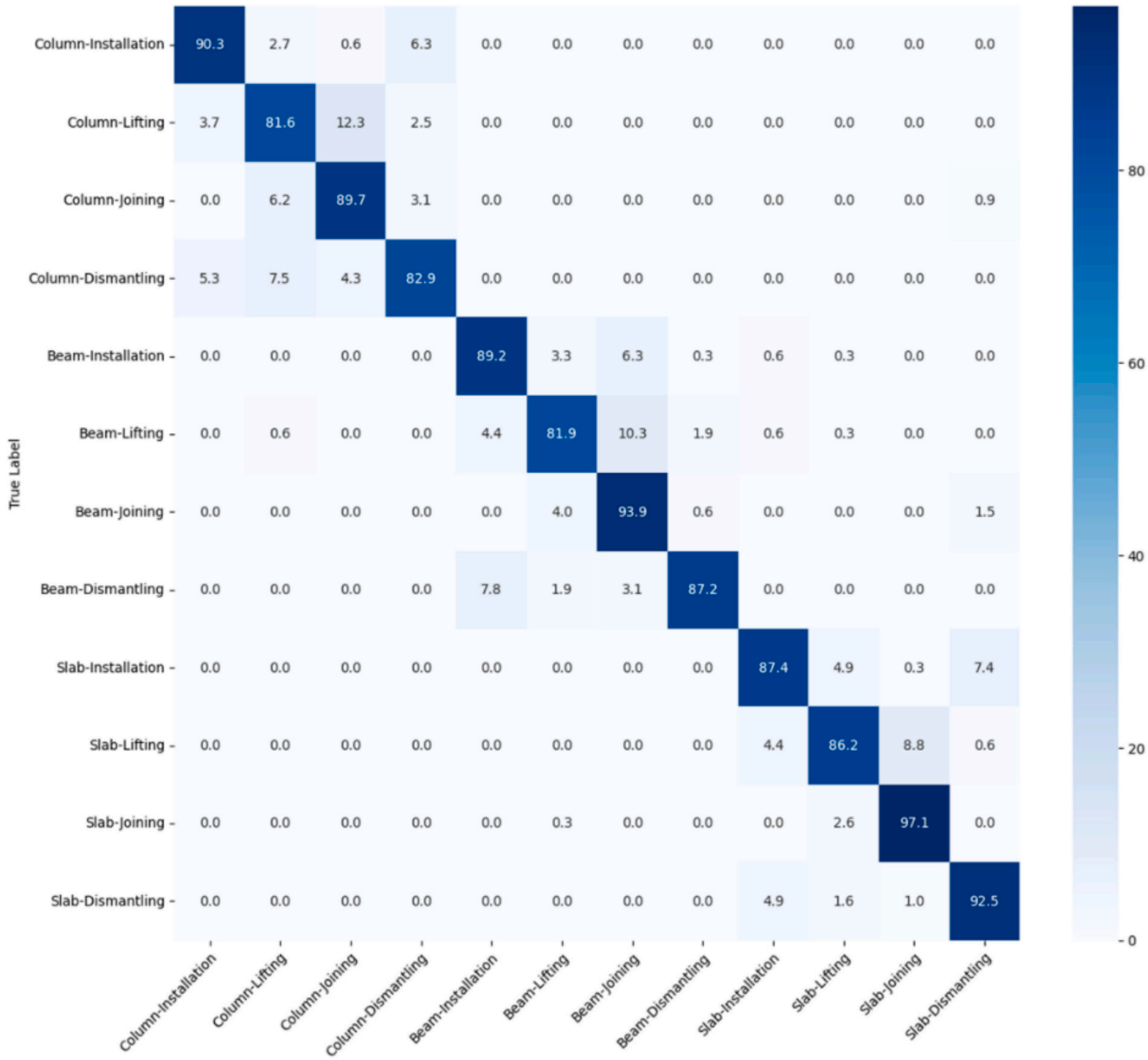

Fig. 5. VideoMAE V2 confusion matrix.

5. Limitations and future research directions

This study has several limitations in automatically recognizing PC component installation actions using the PCI-Dataset. First, the model's generalization ability may be limited due to the need for scale and diversity of the dataset. Also, the limited scope of installation action annotations may limit the ability to recognize some critical installation action stages or particular situations. The fact that video data's temporal continuity and context were not sufficiently considered may also affect the accurate recognition of continuous installation action processes. Fourth, this study's focus on video data alone may not capture all aspects of the complex PC installation process, potentially limiting the model's comprehensiveness. Lastly, the possibility of errors in the manual annotation process may act as noise during model training, potentially degrading overall recognition performance.

To overcome these limitations, the following future research directions are proposed: The scale of PCI-Dataset should be expanded, and its diversity should be increased to improve the model's generalization ability. Developing new model structures that can better reflect the sequential characteristics of PC component installation actions is required. The introduction of multimodal learning methods integrating voice, sensor, and video data should be considered to provide a more comprehensive understanding of the PC installation process. Additionally, integration with existing datasets such as SODA could enhance the model's ability to understand the broader context of construction sites. The introduction of multimodal learning methods that integrate voice and sensor data in tandem with video data should be considered. Real-time processing and feedback systems should be developed to increase practical usability in the field. Methods to achieve high performance with limited data by applying transfer learning and meta-learning techniques should be researched. Integration plans with existing construction management systems should be studied to explore ways to effectively utilize automatic recognition results in actual process management and decision-making.

Through these future studies, it is expected that the automatic recognition technology for PC component installation actions will be further advanced, and its applicability and effectiveness on actual construction sites will be significantly improved. By addressing the current limitations and expanding the scope of data and methodologies, we aim to develop a more comprehensive and practical tool for PC installation management, contributing to the broader goal of digitizing and optimizing construction processes.

Fig. 6. Per-installation action performance comparison of VideoMAE V2, TimeSformer, and ViViT models for PC component installation recognition.

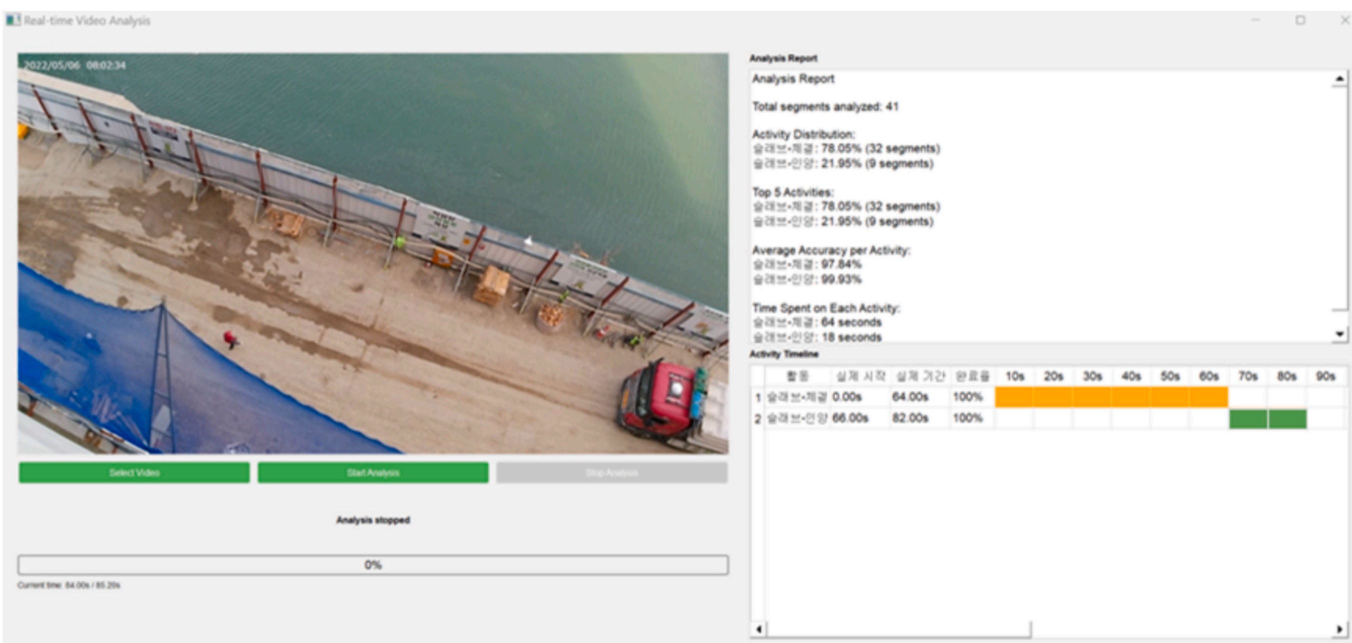

Fig. 7. Installation action recognition prototype system for a case study.

6. Conclusions

This paper constructed a PCI-Dataset for automatic recognition of PC component installation actions, evaluate the performance of Transformer-based video recognition models using this dataset, and verify the effectiveness of the developed model by applying it to an actual PC component installation site. The main results and significance of the research can be summarized in detail as follows:

First, this study successfully constructed a PCI-Dataset consisting of 12,791 video clips collected from actual PC construction sites. This dataset includes 12 installation action classes combining three PC components (column, beam, slab) and four installation action stages (rigging, lifting, assembly, and unrigging). Each class is composed of a balanced proportion between 8.1 % and 8.5 % of the entire dataset, designed to prevent the model from being biased towards specific components or installation action stages. Also, reflecting the variability of actual installation environments, including weather conditions, time zones, worker experience, and equipment types, is expected to improve the model's generalization ability.

Second, this study compared and evaluated the performance of Transformer-based video recognition models such as VideoMAE V2, TimeSformer, and ViViT by applying them to PCI-Dataset. As a result, the VideoMAE V2 model showed the best performance with 98.10 % accuracy, while ViViT and TimeSformer recorded 88.45 % and 87.76 % accuracy, respectively. The excellent performance of the VideoMAE V2 model is interpreted to be due to the contribution of Masked

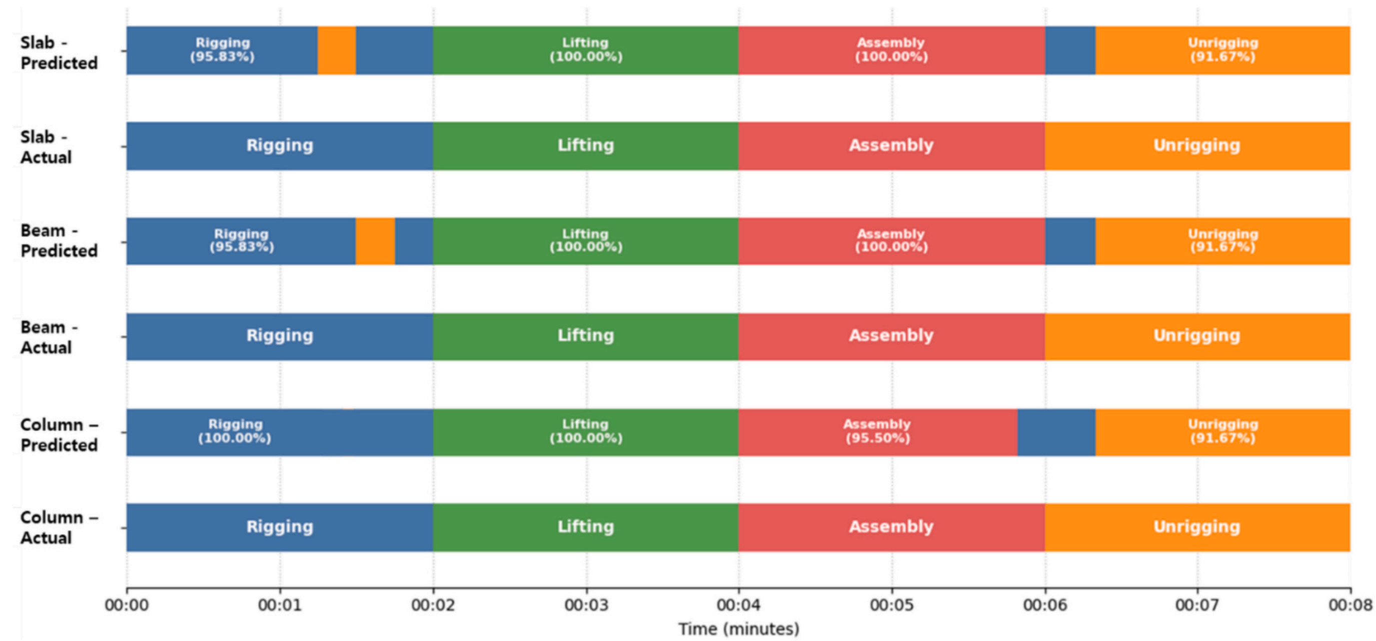

Fig. 8. PC component installation work recognition results of VideoMAE V2 model.

Table 11
Recognition analysis results of PC component installation work over seven days.

Categories	Activity	Recognition Accuracy (%)	Total Time (hour)	Average Time Ratio (%)	Average Time per Unit (min)	Average Wait Time (*sec*)	Productivity (units/h)
Beam	Rigging	98.4	10.7	4.71	10.7	100.8	5.6
	Lifting	97.1	22.9	10.0	28.6	286.8	2.1
	Assembly	98.1	33	14.5	40	415.8	1.5
	Unrigging	98.2	3.7	1.6	7.3	43.8	8.2
Slab	Rigging	98.7	2.6	1.1	2.6	25.2	23.1
	Lifting	98.1	18.7	8.2	18.8	175.8	3.2
	Assembly	98.1	13.3	5.8	13.3	124.8	4.5
	Unrigging	98.7	1.4	0.6	1.4	13.2	43.6
Column	Rigging	97.7	16.6	7.3	16.7	142.8	3.6
	Lifting	97	39.1	17.2	40	337.2	1.5
	Assembly	98.4	51.3	22.5	54.5	442.2	1.1
	Unrigging	97.6	13.9	6.1	14.3	120	4.2
Total		98.1	227.2	100	20.6	185.7	8.5

Autoencoder-based pre-training to effective feature representation learning. In particular, the VideoMAE V2 model showed high recognition accuracy of over 80 % for most installation actions and achieved accuracy exceeding 90 % for actions such as beam-assembly, beam-unrigging, and slab-assembly.

Third, this study verified the effectiveness of the developed model by applying it to an actual PC construction site over a 7-day period. The Value Stream Map analysis of installation processes revealed distinct temporal patterns across components ($n = 48$ beams, $n = 52$ slabs, $n = 42$ columns). For beam installations, assembly operations showed the highest variability (36.0–44.0 min), while slab installations demonstrated higher efficiency, averaging about 36.1 min. Column installations required the longest time, averaging about 125.5 min. Further analysis of the entire week's data confirmed the model's consistent performance, showing recognition accuracies ranging from 90.4 % to 98.7 % across all installation stages. This extended analysis also provided valuable insights into task distribution, productivity, and potential bottlenecks in the installation process. These results demonstrate that the developed model can maintain high performance despite the complexity and diversity of real construction sites, and can provide actionable insights for construction management over extended periods. Integration of digital technologies (e.g., sensors, BIM, automated equipment monitoring) with these temporal insights enables enhanced decision-making and situational awareness, particularly in predicting and preventing excessive waiting periods through real-time monitoring and automated equipment tracking.

This study contributes to expanding construction management theory by applying computer vision technology to OSC project management. It would enable real-time and automated installation action recognition and analysis on fields, which is totally different from the existing passive and retrospective project monitoring methods. By introducing deep learning-based video recognition models to OSC project management theory, it has laid the foundation for promoting the digitalization and automation of OSC sites and contributing to improving project efficiency and safety. This technology presents potential for construction automation: real-time quality control, predictive maintenance, and safety monitoring through action analysis. The integration of these capabilities with existing digital technologies (e.g., BIM, IoT sensors) could enable data-driven decision making and problem resolution. This suggests a transition from traditional human-centered supervision to AI-assisted construction management, where the roles of construction professionals and requirements for new technical capabilities are likely to evolve. As construction sites become more digitalized, the industry may need to balance automation benefits with human

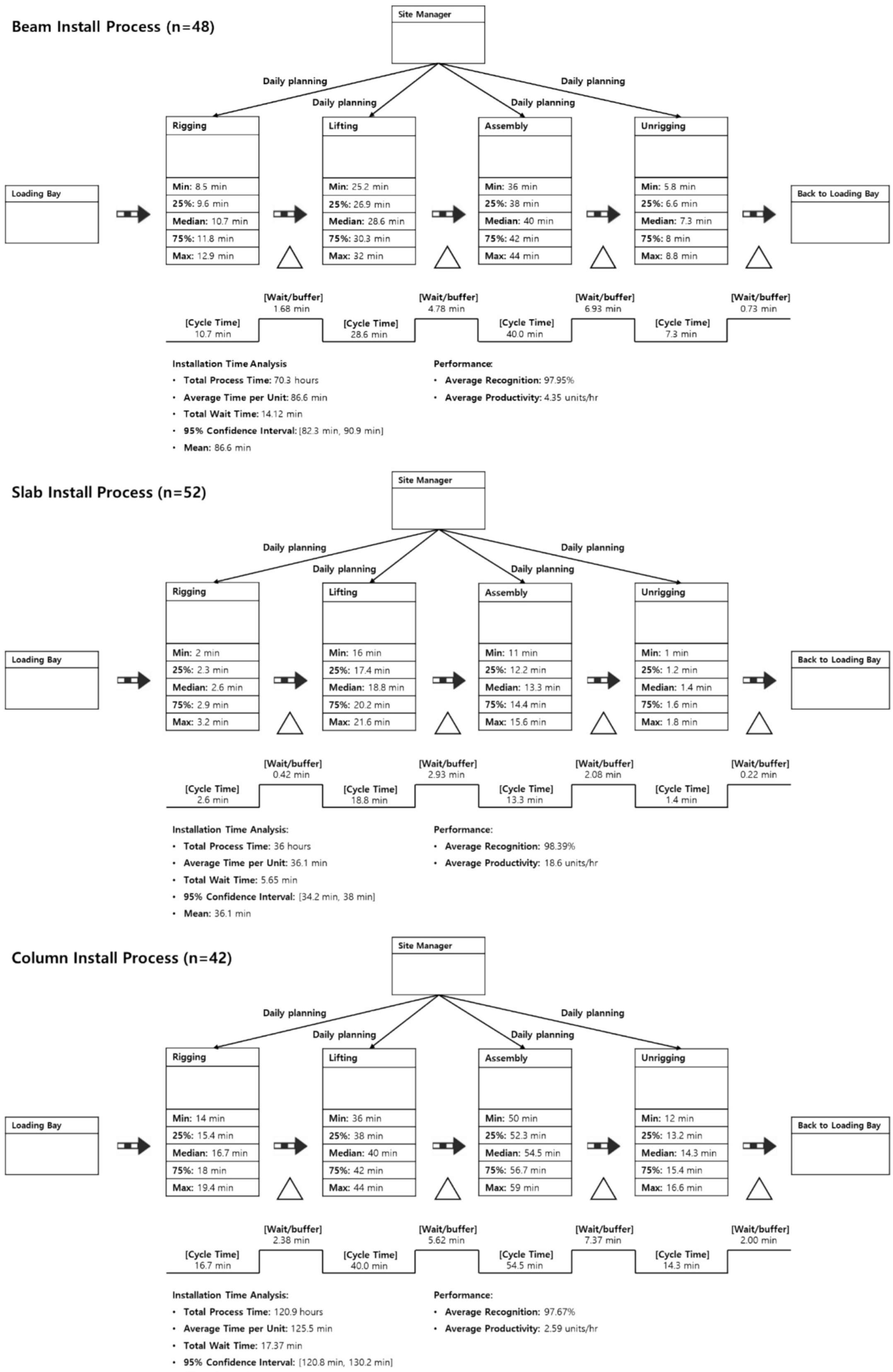

Fig. 9. Value stream mapping of beam, slab, and column installation processes with time distribution analysis.


Fig. 10. Misrecognition cases in actual site application.

judgment in critical decision-making processes.

CRediT authorship contribution statement

Junyoung Jang: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Eunbeen Jeong:** Visualization, Validation, Formal analysis, Data curation. **Tae Wan Kim:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Some or all of the data or code that support the findings of this study are available from the following links on Laboratory of Construction Engineering and Management, Incheon National University (http://inucem.inu.ac.kr) and GitHub(URL to be added).

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